

Core Definitions & Linkage Equations

Eq. #	Mathematical Relation	Description / Role
(15a)	$f_m(Ri) = \phi_m^{-2}(\zeta)$	Definition of momentum stability function f_m
(15b)	$f_h(Ri) = \phi_m^{-1}(\zeta)\phi_h^{-1}(\zeta) = \frac{f_m^{1/2}}{\phi_h(\zeta)}$	Definition of heat stability function f_h
(16)	$Ri = \zeta \frac{\phi_h(\zeta)}{\phi_m^2(\zeta)}$	Richardson number expressed via MOST profile functions
(17)	$\zeta = Ri \frac{f_h(Ri)}{f_m^{3/2}(Ri)}$	Corrected ζ - Ri transformation equation

Stable Boundary Layer ($\zeta > 0$)

Eq. #	Mathematical Relation	Description / Role
(5a)	$\phi_m = 1 + \beta_m \zeta, \quad \phi_h = (1 + \beta_h \zeta)^{\alpha_h}$	Linear/generalized stable MOST profile functions
(23)	$\zeta = \frac{1-x}{\beta_m x}, \quad \text{where } x \equiv f_m^{1/2}$	Inverted profile mapping ζ to $f_m^{1/2}$
(24)	$Ri(\beta_h - \beta_m)x^2 - (1 + Ri\beta_h)x + 1 = 0$	Master quadratic equation for $\alpha_h = 1$
(25)	$f_m(Ri) = x^2 = \left[\frac{(1+Ri\beta_h) - \sqrt{(1+Ri\beta_h)^2 - 4Ri(\beta_h - \beta_m)}}{2Ri(\beta_h - \beta_m)} \right]^2$	Exact closed-form solution for stable $f_m(Ri)$

Unstable Boundary Layer ($\zeta < 0$)

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Eq. #	Mathematical Relation	Description / Role
(33a)	$f_m(\zeta) = (1 - \beta_m \zeta)^{1/2}$	f_m from Businger–Dyer profile ($\alpha_m = -1/4$)
(33b)	$f_h(\zeta) = (1 - \beta_h \zeta)^{1/2} (1 - \beta_m \zeta)^{1/4}$	f_h from Businger–Dyer profile ($\alpha_h = -1/2$)
(35)	$f_h(f_m) = f_m^{1/2} \left[1 - \frac{\beta_h}{\beta_m} (1 - f_m^2) \right]^{1/2}$	Direct coupling relation expressing f_h through f_m
(36)	$x^3 - 2x^2 + (1 - Ri^2 \beta_m \beta_h)x - Ri^2 \beta_m (\beta_m - \beta_h) = 0$	Unstable master cubic in $x \equiv f_m^2$
(37)	$\phi_m(Ri) \approx (1 - 16Ri)^{-1/4}$	Direct Ri -dependent profile function approximation
(38)	$f_m(Ri) = (1 - 16Ri)^{1/2}$	Closed-form solution for $f_m(Ri)$ when $\beta_m = \beta_h = 16$